

A PROJECT TITLE
ON
**<Multidimensional Soil Nutrient Data Visualization and Feature Dimensionality Reduction
Using Principal Component Analysis>**

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
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by

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DEPARTMENT OF Computer Science and Engineering

Name of Title: Multidimensional Soil Nutrient Data Visualization and Feature Dimensionality Reduction Using Principal Component Analysis

1. Abstract:

Modern precision agriculture relies heavily on analyzing complex, high-dimensional soil health parameters—such as Nitrogen (N), Phosphorus (P), Potassium (K), pH, organic carbon, electrical conductivity, and micronutrients—to optimize yield and fertilizer usage. However, analyzing and interpreting these correlated soil properties across multiple geographical zones presents significant visualization and analytical challenges. High-dimensional feature spaces often suffer from multicollinearity and visual opacity, making it difficult for agronomists and machine learning models to extract actionable insights. This project implements Principal Component Analysis (PCA) to compress high-dimensional soil nutrient datasets into lower-dimensional feature spaces (specifically 2D and 3D principal components) while retaining maximum dataset variance. By normalizing soil parameters using Z-score standardization and computing eigenvectors and eigenvalues from the feature covariance matrix, the system transforms raw physical measurements into orthogonal, uncorrelated components. The resulting principal components are visualized using interactive biplots, scree plots, and cumulative variance distribution charts. The primary objective is to visually cluster soil profiles based on nutrient similarity, identify spatial soil degradation, and isolate key driving variables through eigenvector feature loading analysis. Results demonstrate that reducing the feature space to two or three principal components preserves over 80% to 85% of the overall variance, enabling clear visual segmentation of soil quality zones without severe loss of information. This dimensionality reduction pipeline serves as an essential exploratory data analysis (EDA) and pre-processing step for downstream agricultural predictive modeling, crop recommendation systems, and precision fertilization strategies.

2. INTRODUCTION:

Agricultural soil testing yields datasets characterized by complex interdependencies among physical and chemical attributes. Key macro-nutrients (N, P, K) and micro-nutrients (Fe, Zn, Mn, Cu) often exhibit strong correlations influenced by soil type, irrigation practices, and climate. Visualizing these relationships directly in higher dimensions is impractical. Unsupervised machine learning techniques, particularly dimensionality reduction algorithms, provide a structured approach to simplify complex agricultural datasets. Principal Component Analysis (PCA) performs a linear orthogonal transformation to project multivariate soil data onto a new coordinate system defined by directions of maximum variance. In this project, PCA is applied to preprocess and visualize multidimensional soil nutrient profiles. The system pipeline consists of: Data cleaning and standard scaling (to account for different units like ppm, %, and pH values). Calculating the covariance matrix to capture inter-variable dependencies. Eigendecomposition to derive Principal Components (PC₁, PC₂, ..., PC_n). Projecting datasets onto 2D and 3D scatter spaces with loading vectors (biplots) to reveal structural clusters, outlier sample regions, and variable correlations. By reducing dimensionality while preserving signal integrity, this approach helps agronomists easily interpret soil spatial variations and optimize nutrient management decisions.

3. SOFTWARE REQUIREMENTS:

- **Python 3.10+:** Core programming language.
- **Jupyter Notebook / Google Colab:** Interactive development environment for model experimentation and visual inspection.
- **NumPy & Pandas:** Data manipulation, matrix operations, and dataset structuring.
- **Scikit-Learn:** Implementing PCA (`sklearn.decomposition.PCA`), StandardScaler (`sklearn.preprocessing.StandardScaler`), and preprocessing routines.
- **Matplotlib & Seaborn:** Static visualization of scree plots, heatmap covariance matrices, and 2D projections.
- **Plotly Express / PyVista:** Interactive 3D scatter plots and biplots for spatial nutrient exploration.

Signature of the Guide

Course Instructor